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An Intelligent Decision Support System for Investment Analysis

K. L. Poh

Department of Industrial and Systems Engineering, National University of Singapore, Singapore

Abstract. We present an intelligent decision support system that combines decision analysis and traditional investment evaluation and analysis. The system brings to the ordinary user expertise in both decision analysis and investment evaluation techniques, as well as domain knowledge about the market. The system supports the entire decision analysis cycle and provides facilities and tools for decision modeling, probability assessment, model evaluation, and sensitivity analysis. An example based on the Shanghai Stock Market is presented. We demonstrate how an investment decision model for the stock market is constructed by the system and how the optimal decision is obtained. Sensitivity and value of information analysis were also carried out by the system.

Keywords: Decision analysis; Intelligent decision system; Investment analysis; Knowledge-based decision support system

1. Introduction

Investment decisions in the real world are complex processes which are often complicated by the vast amount of information as well as the high level of risk involved. Hence, in developing any automated investment decision support system or tool, the representation and management of this vast amount of information must be carefully and effectively addressed so that sound and timely decisions may be made by the investors. This paper presents an intelligent decision support system for investment decision making in which normative decision theory and decision analysis (Howard, 1996a; Raiffa, 1968; Savage, 1972; Clemen, 1996) are combined with traditional investment analysis and evaluation techniques (Reilly and Brown, 1997) to provide an integrated intelligent decision support

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environment. The system brings to the user expertise in decision and investment analyses as well as domain knowledge of specific financial markets.

The management of investments in the global financial market has recently experienced a lot of changes with advances in information and communication technology. In the global stock markets, almost all stock exchanges in the world have become either fully or partially computerized. This resulted in a number of significant changes in the market. First, with electronic trading, transaction volumes have increased tremendously and transaction times have significantly shortened. For example, in 1992, the world's oldest and largest trading system, INSTINET, reported that its volume had doubled within a relatively short period, while the Barra's Portfolio System for Institutional Trading (POSIT) reported a 50% increase (Winninghoff, 1993). Second, advances in information and communication technology have provided investors with access to the global markets, and an increasing number of stocks have secondary listings on foreign stock exchanges. Finally, with the vast amount of information generated from the financial markets, companies such as Reuters and Standard & Poor, which specialize in the sales of market information, have emerged.

Investment managers operating in the dynamic market must have the ability to extract relevant information in the shortest possible time and identify key variables which have significant impact on the return to investment in order to make timely decisions. Researchers in portfolio management have developed models which attempt to value assets and estimate the expected return of securities or portfolios. Two well-known models are the Capital Asset Pricing Model (CAPM) (Sharpe, 1964) and the Arbitrage Pricing Theory (APT) (Ross, 1976). While these two models are widely accepted, their applications require the user to be formally and adequately trained in investment theory and financial management.

Our system combines decision analysis with traditional investment evaluation approaches. The system has an architecture which can be considered as a form of Klein and Methlie's knowledge-based decision support system (KB-DSS) (Klein and Methlie, 1990, 1995) as well as Holzman's intelligent decision system (IDS) (Holtzman, 1989). As a KB-DSS, the system brings to the user expertise in both decision and investment analyses as well as domain knowledge of specific financial markets. In particular, the knowledge required to perform the decision analysis cycle (Matheson and Howard, 1968) as practiced by professional decision analysts is supported by the system together with investment analysis methodologies. As an IDS, the system dynamically builds decision models using influence diagrams according to the user's inputs, and does not assume a 'fixed' or 'static' decision model structure for all situations. However, unlike Holtzman's original IDS, our system does not employ a rule-based inference engine for decision model construction. Instead, our system employs a model construction procedure that will dynamically build a comprehensive influence diagram representing the situation-specific decision problem which includes factors that are considered in well-established financial analysis procedure.

This paper is organized as follows: In Section 2, we provide a review of related work on investment analysis, portfolio management, and decision analysis. In Section 3, the system architecture and the various subsystems and modules are described. In Section 4, we show how the system may be applied to solve an investment decision problem related to the Shanghai Stock Market. Finally, in Section 5 we conclude with a discussion on the system's limitations and suggest possible extensions to the system.

2. Investment Analysis, Portfolio Management, and Decision Analysis

Investment may be defined as the current commitment of financial resources or funds for a period of time in order to derive future payments that will compensate the investor for the time the funds were committed, the expected rate of inflation, and the uncertainty of the future payments (Reilly and Brown, 1997). Most investments have expected cash flows and a stated market price. In evaluating the value of an investment, an investor considers the investment's expected cash flows and the required rate of return. The estimated values of the securities are then compared to their market prices before a decision is made on which securities to buy. Reilly and Brown (1997) proposed a three-step evaluation process for investment analysis as follows:

1. Analysis of alternative economies and security markets
2. Analysis of alternative industries
3. Analysis of individual companies and stocks

The first step is concerned with the analysis of security markets which will reflect the state of the economy. The value of an investment is determined by its expected cash flows and its required rate of return. Both of these factors are strongly influenced by the state of the economy. For example, the earnings of a firm operating in a strong and fast expanding economy is expected to increase, and the investor's rate of return on the company stock is expected to be high. On the other hand, if the economy is in recession, the earnings of the same firm would not be expected to increase in the short run and its stock price would be either stable or declining. Therefore, when assessing the value of a security, it is necessary to analyze the aggregated economy together with the security markets.

The second step is concerned with industry analysis, which is important for a number of reasons. First, returns may vary widely over any time period. Second, the rates of return for individual industries may also vary over time. Hence investors cannot simply extrapolate past industry performances into the future. Thirdly, risk levels for different industries may vary widely during any time period. Industry analysis can therefore help investors isolate profitable investment opportunities from those that are non-profitable.

The third and final step is concerned with company analysis. This step helps the investors find opportunity stocks whose values are under-priced. Company analysis is separated from stock analysis because the stock of a good company is not necessarily a good investment. For example, the stock of a good firm with superior management measured by its current sales and earnings growth may be over-priced such that the actual value of the stock is below its market value. On the other hand, the stock of a company with less success based on its current sales and earnings growth may have a stock market value that is below its intrinsic value. In this case, although this company is not as good as the first, its stock could be a good addition to the investor's portfolio. Hence, it is necessary to distinguish between growth companies and growth stocks. A growth company is one with an able management that is capable of identifying opportunities to make investments that yield rates of return greater than the firm's required rate of return. A growth stock, on the other hand, is one with a rate of return that is higher than other stocks in the market with similar risk characteristics (Ezra, 1963).

Portfolio management deals with the selection of portfolios that maximize expected returns at some acceptable levels of risk (Fabozzi, 1995). Markowitz (1952, 1959) laid the foundation for modern portfolio analysis and provided for the first time a systematic framework for selecting securities and portfolios on the basis of their expected returns and risks. He also pioneered the measurement of security and portfolio risk in terms of the well-known statistical measure known as the variance. The variance of a random variable is a measure of the dispersion of the possible outcomes around its expected value. Markowitz defined the variance of a portfolio as the sum of the weighted variances of the individual assets plus the sum of the weighted covariances of the assets. Based on the expected values and variances of possible portfolios, optimization techniques can be used to determine the Markowitz efficient portfolio. A Markowitz efficient portfolio is one which has the highest expected return among all feasible portfolios with the same level of risk. The Markowitz efficient frontier is then the collection of all Markowitz efficient portfolios at different levels of risk. The investor's utility is then maximized when the indifferent utility curve is tangential to the Markowitz efficient frontier.

Based on Markowitz's portfolio theory, the Capital Asset Pricing Model (CAPM), which attempts to estimate the expected rate of return to securities or portfolios, was developed (Sharpe, 1964, 1985; Litner, 1965). CAPM assumes that the expected return to any security or portfolio is determined by its relationship with a special portfolio known as the market portfolio. This is based on the observation that the returns to most securities or portfolios tend to move together with the returns to the market portfolio. The market portfolio consists of all assets available to investors, and each asset must be held in proportion to its market value relative to the total market value of all assets.

The CAPM model has been severely criticized for its dependence on the existence of a market portfolio which cannot be observed in practice. This is because such a portfolio contains all the assets in the economy, and an asset or a portfolio's expected return is dependent on the sensitivity of the asset or portfolio's return to the excess returns on more than one factor. In 1976, another model, called Arbitrage Pricing Theory (APT), was developed (Ross, 1976). The APT model seems to be better in explaining the apparent puzzle as to why the estimated returns to securities issued by small firms tend to be higher than the estimated expected returns to securities issued by large firms (Sivalingam, 1990). Although the APT suffered from difficulties in identifying the required factors, it required fewer assumptions and considered multiple factors to explain the risk of an investment, in contrast to the single-factor CAPM.

Decision analysis (Howard, 1996a; Raiffa, 1968; Clemen, 1996) is a term used to describe a body of knowledge and professional practice for the logical illumination of decision problems. Decision analysis has a history of more than 30 years and was developed from statistical decision theory (Savage, 1972), operations research and systems analysis. Decision analysis seeks to apply logical, mathematical, and scientific procedures to complex decision problems through mathematical modeling and structuring of the decision situation, computational implementation of the model and, finally, quantitative evaluation and comparison of the alternative courses of action (Matheson, 1968). Much of the power of decision analysis lies in its ability to integrate effectively many factors that affect a decision, and to clearly and concisely represent decision problems which are necessary for helping decision-makers gain insight into the problems (Clemen, 1996). Decision analysis has been applied to a wide range of problems including plant expansion planning

(Braunstein, 1983), new product development (Matheson, 1970), space program planning (Matheson, 1970), investment in water supply systems (Rabinowitz et al, 1988), selection of residence (Hoepfner and Mata, 1993), energy investment (Huang et al, 1995, 1997), facility investment (Spetzler and Zamora, 1983), etc.

In general, decision analyses are expensive and time consuming to perform. Usually, a decision analysis needs one or more decision-makers, one or more domain experts, and one or more decision analysts (Heckerman, 1991). Much of the cost of building a decision model is in capturing the expert knowledge into the model. In recent years, researchers have attempted to reduce the high cost of decision analyses by providing intensive domain knowledge in computer-based decision systems based on normative decision theory. Examples of such systems are the normative expert systems (Heckerman, 1991) and the intelligent decision systems (Holtzman, 1989). The latter have been identified by Klein and Methlie (1990, 1995) as special forms of knowledge-based decision support systems.

Normative and intelligent decision systems differ from other traditional decision support systems in that they combine the knowledge-based paradigm from artificial intelligence for domain knowledge representation, and normative decision theory and analysis for inference. It should also be noted that these systems have been realizable only recently due to the development of an efficient decision modeling and inference language, namely the influence diagram and coupled with efficient probabilistic reasoning techniques for building automated decision aids and computer-based systems (Miller et al, 1978; Owen, 1978). Influence diagrams (Howard and Matheson, 1984) are *directed acyclic graphs* (DAG) for expressing and representing decision models. The nodes in an influence diagram represent the variables or events that are relevant to the decision problem, while the arcs represent the relations between the variables. Influence diagrams are compact and easy to manipulate by computers, unlike the traditional decision trees which grow exponentially in size for large problems. Furthermore, influence diagrams are capable of explicitly representing probabilistic dependencies, in contrast to the decision trees, which cannot do so.

In a normative expert system, the domain knowledge and decision model are typically pre-engineered and represented as an influence diagram. Inference is then performed on the influence diagram and the structure or topology of the diagram does not change very much. An intelligent decision system, on the other hand, does not have a decision model pre-built. Instead, general domain knowledge is captured in a knowledge base and a customized situation-specific decision model in the form of an influence diagram is built during run time. Recent research has focused on the representation of domain knowledge and techniques for decision model construction (Breese et al, 1991).

In our system, we combine decision analysis with traditional investment analysis and portfolio management. Traditional investment evaluation techniques do not explicitly consider the risk attitudes of investors. Some investors may be risk averse and their degrees of risk aversion can be measured using risk tolerances. Traditional investment analysis also does not consider probabilistic dependencies among key variables.

Our system attempts to overcome the limitations of traditional investment analysis by modeling the investment decision-making problem at three levels corresponding to Reilly's three-step evaluation approach described earlier. The three levels of analysis are economy and security analysis, industry analysis, and company analysis. Each level of analysis progressively tries to capture the variables which are relevant to the problem, and at the same time enables

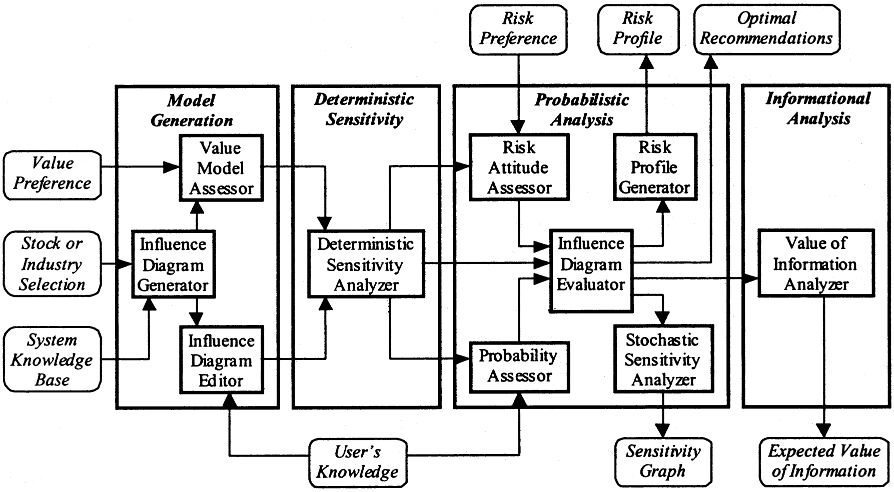

Fig. 1. The system architecture.

the probabilistic dependency relations among all the variables to be captured in the form of an influence diagram. Conditional probability distributions for all uncertain variables are assessed and the risk preferences and attitude of the decision-maker are captured accurately in a utility function. Based on the constructed model, the system can determine the optimal decision policy as well as the optimal expected return. In addition, sensitivity analysis can be performed.

3. System Architecture and Implementation

Figure 1 shows the architecture of the system, which comprises four major subsystems, namely the model generation subsystem, the deterministic sensitivity subsystem, the probabilistic analysis subsystem, and the informational analysis subsystem. Each of these four subsystems corresponds to a distinct phase of the decision analysis cycle (Howard, 1983). The system was developed with the Microsoft Visual C++ development studio using standard Windows-based dialog boxes and interfaces.

It should be noted that many of the features provided by our system can also be found in industry-standard software such as *DPL*, *DATA* and *Precision Trees* for professional decision modeling and analysis. These include model editing functions, sensitivity analysis tools, risk profiles generator, and value of information analyzer. However, these commercial software packages are generic decision analysis tools for professional decision analysts and do not provide any intelligent guidance for building situation-specified models, nor do they provide any specific domain knowledge for the building decision models.

3.1. The Model Generation Subsystem

The model generation subsystem is concerned with creating a decision model based on inputs from the decision-maker. Inputs to this subsystem include the name of the stock market or the industry to be analyzed. Decision models in the system are represented as influence diagrams. The model generation subsystem is further made up of three sub-modules. They are the influence diagram generator, the influence diagram editor, and the value model assessment sub-modules.

The influence diagram generator sub-module generates a preliminary influence diagram comprising three levels corresponding to stock market analysis, industry analysis, and company analysis. The three-level structure of the influence diagram allows the decision-maker to concentrate on developing the model on a level-by-level basis, and avoids being overloaded with a large amount of information. A user may select any one of the three levels of analysis from a dialog box. Alternatively, the user can choose to create an entirely new model without the help of the domain knowledge base.

The influence diagram editor sub-module allows the user to customize the decision model by directly editing the model via a set of influence diagram operations. These operations include addition and deletion of nodes and arcs from the influence diagram. If a decision node is added, the user is asked to specify the alternatives corresponding to the newly added node. If a chance node is added, the user is asked to specify the set of possible outcomes or states for the new node. In addition to providing influence diagram editing facilities, this sub-module also provides files management functions for saving and loading of models from external storage devices.

Finally, the value model assessment sub-module determines the outcomes and consequences that the decision-maker cares about. Preferences and value trade-off are the core of any decision problem. The final decision recommended by the system ultimately depends on the values and preference trade-offs specified by the decision-maker. Typically, each alternative can produce many possible outcomes and these outcomes are subject to uncertainties. The fundamental aim of value assignment is to specify the preferences of the decision-maker for each outcome compared with the others. In our system, the value model is based on the profit return of investment. The value model assessment sub-model helps to identify the state variables that have direct impact on profit return. In our system, the decision-maker either assigns the profit values directly to each combination of the outcomes for the state variables, or he/she can specify a multi-attribute value function.

3.2. Deterministic Sensitivity Subsystem

The deterministic sensitivity subsystem is concerned with deterministic sensitivity analysis of the decision model. Its main aim is to reduce the decision model generated by the model generation subsystem to a manageable size by eliminating unimportant variables or variables that have insignificant impact on the outcomes of the problem. The deterministic sensitivity analyzer is the main module in this subsystem. For each uncertain variable, we let its value vary from its low to high value while keeping all the other uncertain variables at their expected values, and the change in profit return is computed. If the change in the profit return for an uncertain variable is large or significant when compared with those obtained from the other variables, then that uncertain variable has a significant impact on the problem and should be retained in the model. Otherwise, if the change in profit is small or insignificant, then the variable concerned will be fixed at

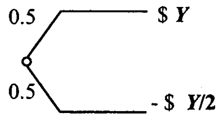

Fig. 2. A reference lottery for assessing risk tolerance.

its expected value in the model for all subsequence analyses. The end result is a smaller model with only the important variables.

3.3. Probabilistic Analysis Subsystem

The probabilistic analysis subsystem is concerned with the probabilistic phase of the decision analysis cycle. Here, all the required probability distributions are assessed, and the influence diagram is then solved for the optimal decision policy. This subsystem comprises five main sub-modules. They are the probability assessment, the risk attitude assessment, the influence diagram evaluation, the stochastic sensitivity analysis, and the risk profile generation sub-modules.

The probability assessment sub-module encodes probability or conditional probability distributions for all the important chance nodes in the model. It also provides tools for assessment of probability distributions for discrete as well as continuous variables.

The risk attitude assessment sub-module determines the decision-maker's attitude towards risk and constructs a utility function for the decision model. Decision theory allows a decision-maker to make rational choices by taking into consideration both returns and attitude to risks. A person's attitude towards risk is characterized by the shape of his utility function over a range of possible value outcomes. A commonly used utility function has the following exponential form:

$$u(x) = 1 - e^{-x/R} \tag{1}$$

where x represents the return and R is a constant known as *the risk tolerance*. Although this utility function is valid only under the condition of constant degree of absolute risk aversion, it is often a good approximation for practice purposes.

Our system provides tools for assessing the decision-maker's risk tolerance. This is performed by posing the reference lottery as shown in Fig. 2 to the decision-maker. Here the decision-maker is asked for the dollar value of y for which he or she is undecided between accepting and not accepting a lottery in which he or she has a 50% chance of winning y dollars or a 50% chance of losing half of y dollars. It can be shown if the decision-maker utility function can be approximated by the exponential form, then the value of y provided by the decision-maker is approximately equal to the risk tolerance R of the decision-maker with an error of only 4% (Howard, 1988).

The influence diagram evaluation sub-module computes the optimal decision policy based on the principle of maximum expected utility. The system uses an extended version of the Shachter node reduction algorithm (Shachter, 1986) to solve the decision model. The extensions include preprocessing of the decision model to eliminate null information nodes in addition to the elimination of barren nodes as originally suggested by Shachter. A null information node is one which does not have any impact on the optimal decision policy. It may be eliminated prior to any node reduction operations to reduce the overall computation time.

The risk profile generation sub-module constructs a risk profile for each of the alternatives so that the risk-return characteristics of the investment alternatives may be compared. The risk profile of an alternative is a plot usually in the form of a cumulative or excess probability curve that shows the probability associated with all possible outcomes of that alternative. Stochastic dominance analysis may be carried out to eliminate alternatives that will never be selected under any valid utility function.

Finally, the stochastic sensitivity analysis sub-module performs probabilistic sensitivity analysis by varying selected model parameters over ranges of possible values. Stochastic sensitivity analysis determines which uncertainty matters in the decision. In our system, any chance node can be selected by the user for stochastic sensitivity analysis. The probability of an outcome for that node can be varied over any range between 0 and 1, and the system will systematically analyze the optimal decision and the maximum expected utility over the specified range of values. The results are presented in a sensitivity graph often called a 'rainbow diagram'.

3.4. Informational Analysis Subsystem

The informational analysis subsystem is concerned the final phase of the decision analysis cycle. The main sub-module here performs expected value of information analyses. When a decision-maker is faced with uncertainty, he or she often has the option of gathering information with the intention of reducing or eliminating the uncertainty so as to improve the decision made. The economic worth of completely eliminating the uncertainty in a chance variable is measured by its expected value of perfect information (Howard, 1966b, 1967). If a variable has an expected value of perfect information that exceeds the cost of acquiring it, then an information-gathering activity may be designed to reduce or eliminate the uncertainty surrounding that variable. The expected value of perfect information of a variable indicates the maximum amount of money one should spend on acquiring information concerning that variable.

The system supports computation of expected values of perfect information as well as expected values of imperfect information. The modeling of the decision problem with perfect information can be represented through appropriate addition of an information arc from the selected chance node to the appropriate decision node. For the imperfect information case, an additional chance connected to the original node is added in the influence diagram to represent the imperfect information that is being observed. An appropriate information arc is also added and the computation is similar to that for the case of perfect information.

4. An Application to the Shanghai Stock Market

4.1. The Shanghai Stock Market

The Shanghai Stock Market was founded on December 19, 1990. At its opening, it had only one trading hall, 46 seats, eight listed companies and 25 members. The total capitalization was only 1.3 billion yuan (US $150 million) and there were only 30,000 investors. At the end of 1993, within three years of its establishment, the Shanghai Stock Market had grown to six trading halls, 2600 seats, 119 listed

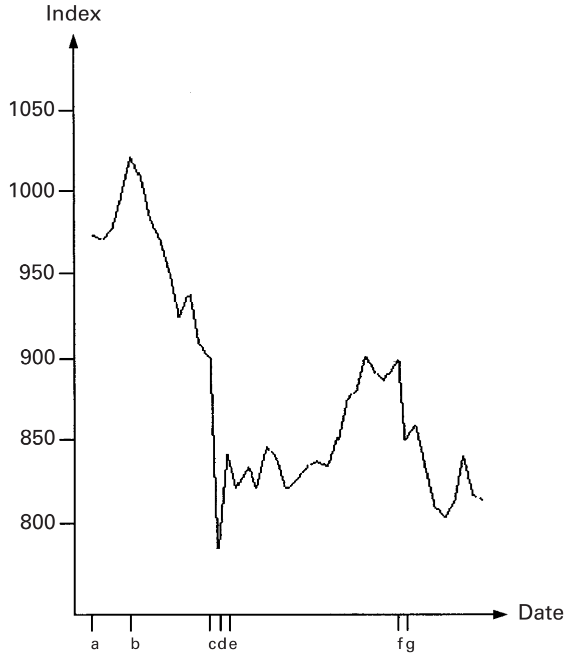

Fig. 3. Index fluctuation from December 1993 to January 1994.

companies, and 481 members. By then, the total capitalization had grown to 250 billion yuan (US $28.7 billion) with 4.6 million investors, and the trading volume was as high as 500 billion yuan (US $57.5 million) (Zhang, 1993).

Our study here focuses on the so-called 'A' shares, which can only be owned by Chinese nationals. The so-called 'B' shares are meant for foreign ownership. Most stock traded as 'A' shares in China are issued in three categories, namely state-owned stock, institutionally owned stocks, and freely traded stocks (Liao, 1994b). In terms of volume, state-owned stocks and institution-owned stocks are much more numerous than freely traded stocks (Liao, 1994a).

The Shanghai Stock Market is still in its early stage of development compared with the mature stock markets in other parts of the world. Some market observers have remarked that the Shanghai Stock Market development trend is contradictory to general economic development trends (Guo, 1994a). At the same time, the Chinese Securities Regulatory Commission (CSRC), being new at its job, is still learning and experimenting with ways of tackling problems that emerged from the stock market. For instance, the *Bao Yan* event in 1993 involving the acquisition of Shanghai-listed *Yanzhong Co. Ltd* by the Shenzhen-listed *Baoan Co. Ltd* caused the price of *Yanzhong* to rise rapidly from about 10 yuan to over 40 yuan per share within a few days. The lack of proper directions and regulatory interventions by the CSRC in the early stage of the acquisition have contributed to the rapid rise in the share price of *Yanzhong*.

The investment awareness of Chinese investors is lower than that of other developed countries. The financial capital already committed in investment of different sorts by the Chinese citizens amounts to only 1.4 trillion yuan (US $161 billion). The amount of bank deposits and cash at hand are as high as 1

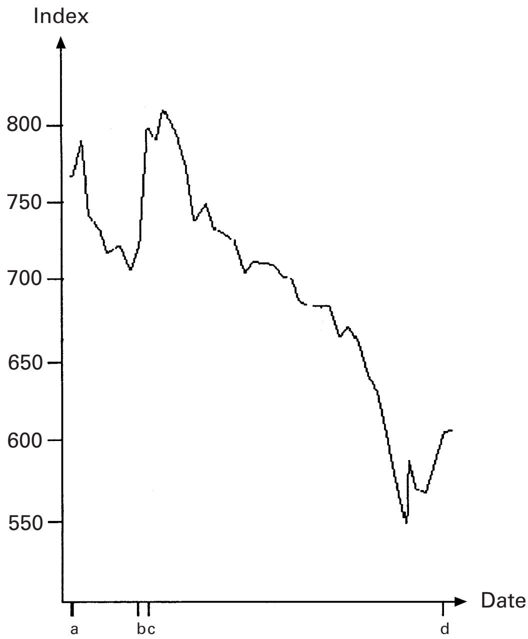

Fig. 4. Index fluctuation from March 1994 to April 1994.

trillion yuan (US $115 billion) and 200 billion yuan (US $33 billion), respectively (Li, 1992). Some stocks have also been noted to change hands up to nearly five times a year due to speculations by some Chinese investors (Liao, 1994a).

The Chinese economic system is in a transition stage from a planned economy to a market economy. State-owned companies operate under a safety net provided by the government, which is a characteristic of the old socialist system. These companies tend to worry less about their longer-term financial health. Some of these companies park a substantial portion of their funds in equities and real estates including the building of infrastructure. The day-to-day operational capital is then raised from the capital market in terms of loans. When the banks' liquidities are driven up and loan facilities are tightened, these companies will find it hard to raise capital for their day-to-day operations (Guo, 1994a). As it is estimated that about up to 80% of the stock market investors are such companies (Liao, 1994b), it is not difficult to understand why the index fell sharply when the Central Government tightened the bank loan policy in 1993, forcing these companies to sell their shares to raise operational capital or to repay their debts (Liao, 1994a, 1994c). Figures 3 and 4 show examples of index fluctuations for the Shanghai Stock Market for the periods from December 1993 to January 1994, and from March 1994 to April 1994, respectively.

4.2. The Decision Model

4.2.1. The Basic Model

A general basic financial investment decision model for an investor in the Shanghai Stock Market is generated by the system as shown in Fig. 5. The domain

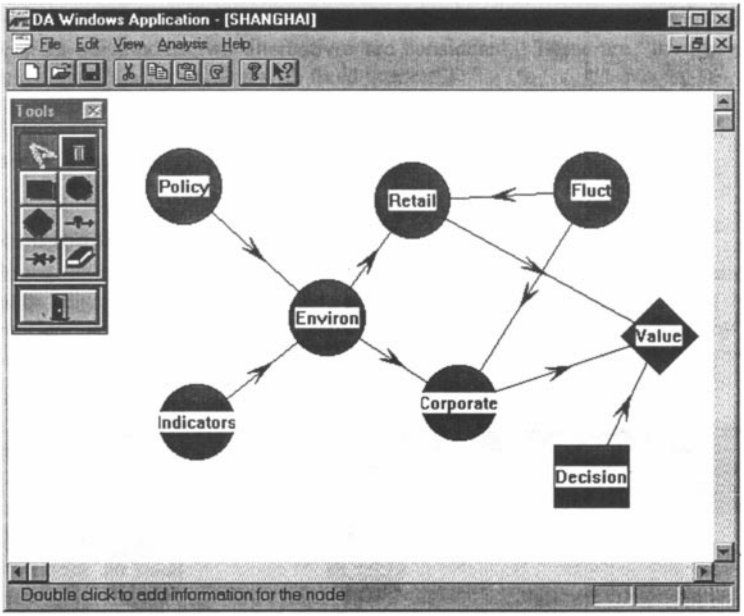

Fig. 5. General financial investment model.

knowledge for the problem was gathered from the literature and publications such as those cited in this paper, and through consultations with financial analysts familiar with the market. We shall describe in detail the various nodes in the influence diagram of Fig. 5.

- Retail investment (*Retail*): This variable reflects the uncertainty of the investment commitments in terms of the trading volume made by individual investors in the stock market. On average, 20% of the trading volume is attributed to retail investment (Liao, 1994b).
- Corporate investment (*Corporate*): This variable reflects the investments made by companies, factories and other state-owned enterprises. On average, 80% of the trading volume is attributed to corporate investment (Liao, 1994b).
- Stock market policy (*Policy*): This variable is concerned with policies introduced by the CSRS. As there are few investment regulations in China, the CSRS will enforce certain stock market policies according to specific situations on an ad hoc basis and these policies usually cause fluctuations of the stock market.
- General economic indicators (*Indicators*): This variable captures the various microeconomic factors which affect both retail investment and corporate investment, such as inflation, interest rate, and GNP growth rate.
- Stock market fluctuation (*Fluct*): This variable denotes the fluctuation of the stock market.
- Investment environment (*Environ*): This is a deterministic variable which is derived from the integration of the 'stock market policies' and 'general economic indicators' chance nodes.
- Investment decision (*Decision*): This decision variable represents the various

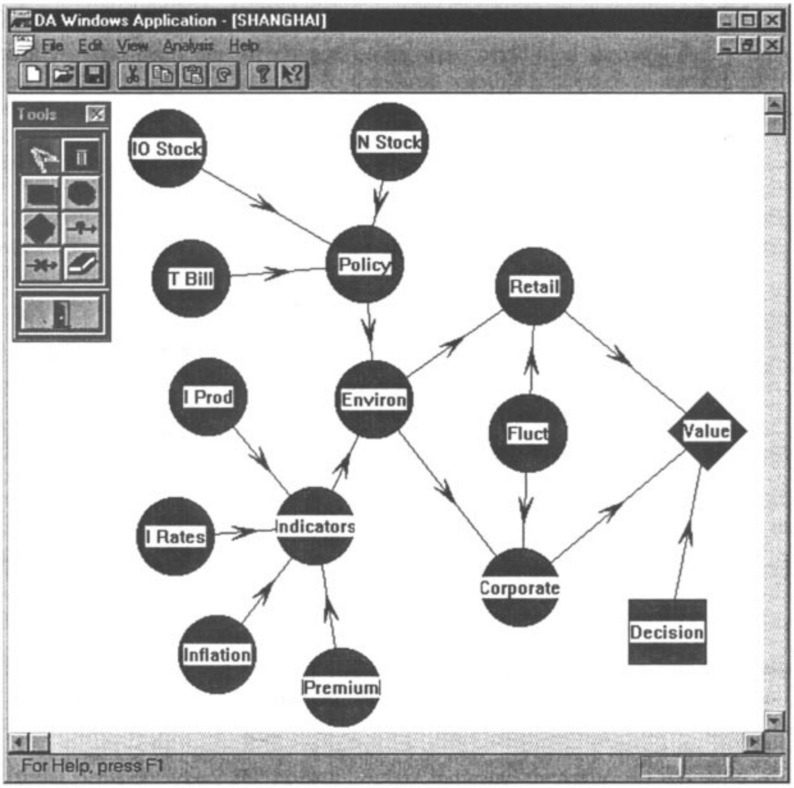

Fig. 6. Decomposition of general decision model.

investment alternatives. In our models two alternatives are considered. These are 'investment in the stock market' and 'saving account with fixed interest'.

- Value (*Value*): This value node represents the profit return and is a function of the uncertain variables retail investment, corporate investment, and investment decision.

4.2.2. Refining the Basic Model

The basic model can be refined by the system by decomposing some of the chance nodes. For instance, Fig. 6 shows how the two chance nodes for general economic indicators (*Indicators*) and stock market policy (*Policy*) of the basic model may be extended by adding a number of direct predecessor nodes.

In the extended model, the general economic indicators (*Indicators*) node is decomposed according to a set of economic variables based on the APT (Rolla and Ross, 1980). The added nodes, which are inflation (*Inflation*), risk premium (*Premium*), interest rates (*I Rates*), and industrial production (*I Prod*), have been observed to have a significant impact on stock market returns. Also in the extended model, the stock market policy (*Policy*) node is extended to include new listings of stocks on the market (*N Stock*), issuing of treasury bills (*T Bill*), and trading of state and institutionally owned stocks (*IO Stock*).

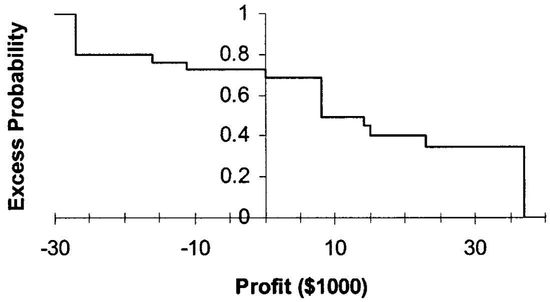

Fig. 7. The risk profile for the optimal decision.

4.2.3. The Value Model

The value model (profit return) is a function of retail investment (*Retail*), corporate investment (*Corporate*), and the investment decision (*Decision*). On average, 80% of the trading volume is due to corporate investment and only 20% to retail investment.

4.2.4. Probability Assessments

Probability assessments for all chance nodes were performed through dialog boxes. For example, probabilities were assigned to the three possible outcomes of stock market policy (*Policy*), namely, *favorable, neutral,* and *unfavorable.* A favorable policy refers to an index-boosting policy where there will be new issues of stocks and treasury bills with properly regulated trading of state and institutionally owned stocks.

Probability distributions for the chance nodes general economic indicators (*Indicators*), investment environment (*Environ*), retail investment (*Retail*), and corporate investment (*Corporate*) were also assessed.

4.2.5. Risk Attitude Assessment

Using the risk attitude assessor, the following lottery was posed to the decision-maker:

- Win $\$Y$ with probability of 0.5.
- Lose $\$Y/2$ with probability of 0.5.

The value of Y for which the decision-maker was just indifferent between accepting and rejecting the lottery was found to be $10,000. Hence, the risk tolerance of the decision-maker is approximately $10,000.

4.3. The Optimal Decision and Risk Profiles

The decision model was solved by an extended version of Shachter's nodes reduction algorithm (Shachter, 1986). After the reduction is completed, only the value node will be left in the influence diagram. The maximum expected value was found to be $10,448.2, and the optimal decision policy was found to be '*investment*

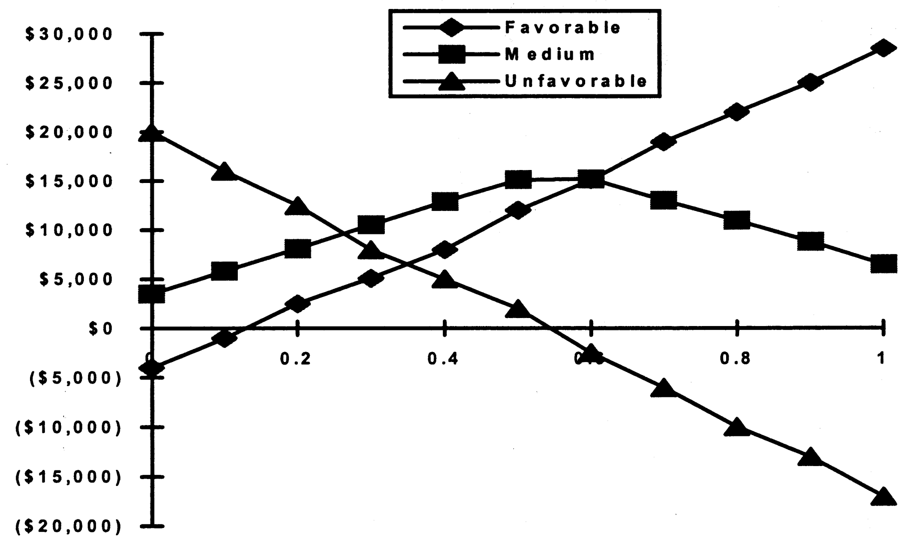

Fig. 8. Probabilistic sensitivity analysis on stock market policy.

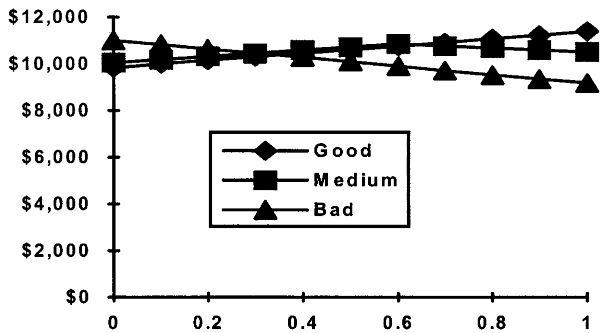

Fig. 9. Probabilistic sensitivity analysis on general economic indicators.

in the stock market'. Figure 7 shows the risk profile for the optimal decision where the horizontal axis indicates the return in thousands of dollars while the vertical axis shows the excess probability, i.e., the probability of achieving at least so much of the value indicated by the horizontal axis.

4.4. Sensitivity Analysis

The decision-maker selected the chance nodes for stock market policy (*Policy*), general economic indicators (*Indicators*), and market fluctuation (*Fluct*) for probabilistic sensitivity analysis. Figures 8–10 show the graphical plots of probabilistic sensitivity analysis on stock market policy, general economic indicators, and market fluctuation, respectively when the probability of the outcome in each variable is varied between 0 and 1. From the sensitivity graphs, we observe that stock

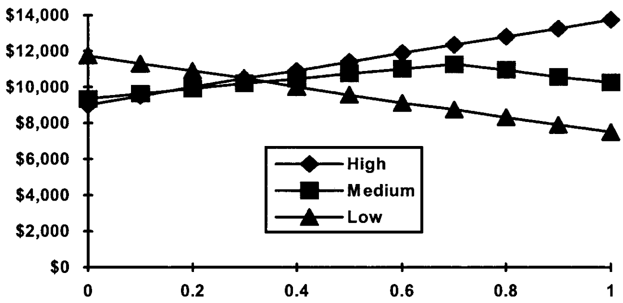

Fig. 10. Probabilistic sensitivity analysis on market fluctuation.

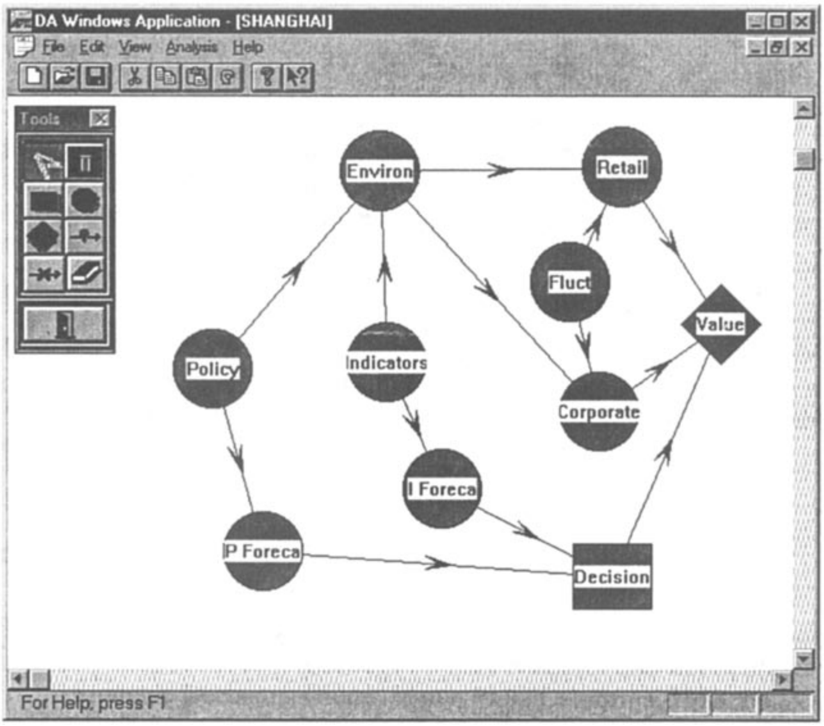

Fig. 11. Expected value of imperfect information analysis.

market policy has the greatest impact on profit return while the general economic indicators have the least impact on profit return.

4.5. Expected Value of Information Analysis

Expected value of imperfect information analysis (Howard, 1996b, 1997) were performed for the chance nodes for stock market policy (*Policy*) and general economic indicators (*Indicators*). The influence diagram for the analysis is shown in Fig. 11. In Fig. 11, two new chance nodes are added to the influence diagram, namely stock market policies forecast (*P Forecast*) and general economic indicators forecast (*I Forecast*), to indicate the imperfect information sources for the two variables, respectively. Conditional probabilities indicating the accuracy or reliability of the information are entered. The modified model was evaluated, yielding a maximum expected value of $14,526.70. Hence the expected value of information is the difference between $14,526.70 and $10,448.20, or $4078.50.

In summary, based on the inputs from the decision-maker, the system has recommended that the optimal policy is to invest in the Shanghai Stock Market with an expected profit return of $10,448.2. It has also shown that stock market policy influences the value (profit return) greatly, while the general economic indicators have little impact on profit return. This is consistent with the observation that the stock market in Shanghai, and in general, China, is a largely policy-influenced market.

5. Conclusion

We have presented an intelligent decision support system that combined decision analysis and theory with traditional investment evaluation procedures to help decision-makers make better investment decisions. The system provides tools such as influence diagram generator, probability assessor, value function generator etc. to help decision-makers model the decision problem. It delivers domain expert knowledge within a decision theoretic framework to improve the quality of decisions. The intelligent decision support system developed here differs from other traditional investment decision and analysis systems in that it takes decision maker's risk attitude and preferences into consideration. Another major feature of the system is the use of influence diagram as a modeling and communication tool for the decision problem.

A limitation of our system is that it cannot help dealers allocate investment funds among the picked stocks such as BARRA's *world markets model*, but it can provide supplementary supports for decision-makers. Although we have shown how the system may be applied to an investment decision problem concerning the Shanghai Stock Market, the system would still require extensive and rigorous testing to prove its worth, and also to assess its portability to other stock markets.

A possible extension of our system is the inclusion of technical analysis (TA). In practice, technical analysis of sentiment and supply and demand through the analysis of actual trading data is often used by investors besides the relative evaluation of the company. We believe that a system that combines both fundamental and technical analyses would be a really useful and powerful one.

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Author Biographies

K. L. Poh is an Associate Professor in the Department of Industrial and Systems Engineering of the National University of Singapore. He obtained his PhD in Engineering–Economic Systems from Stanford University, and is the current President of the Operational Research Society of Singapore. He serves on the Editorial Board of the *Asia–Pacific Journal of Operational Research* and is a member of the Program Committee for the Annual Conference on Uncertainty in Artificial Intelligence.

Correspondence and offprint requests to: K. L. Poh, Department of Industrial & Systems Engineering, National University of Singapore, 10 Kent Ridge Crescent, Singapore 119260. Email: ise-pohkl@nus.edu.sg